PHENIKAA UNIVERSITY
SCHOOL OF INFORMATION SYSTEMS

COURSE CSE703095
SOFTWARE REQUIREMENTS

SAMPLE REPORT – LAB 06

Non-functional Requirements

Topic: Illustrative example of a completed report (reference only, do not copy)

Case study project: MedBook – Online Medical Appointment Booking System
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1 NFR-001, NFR-007 Simulation Results

```
## NFR-001 -- Doctor search performance
- Simulation result: p50=835ms, p95=1233ms, max=2459ms (n=500
   requests)
- Conclusion: PASS (threshold p95 <= 2000ms)

## NFR-007 -- Booking conflict detection safety
- Simulation result: p99=0.824s (n=200 trials)
- Conclusion: PASS (threshold p99 <= 1.0s)
```

Listing 1: nfr_validator.py output (seed=42)

2 Why a Fixed Seed Matters

Setting `random.seed(42)` ensures the simulation is *reproducible* – a basic requirement of scientific/engineering experiments: someone else running the same script must get the same numbers, enabling independent verification (peer review). Without a fixed seed, every run produces different results, making it hard to compare before/after a system optimization.

3 Self-chosen NFR: NFR-002 – Availability

Simulation scenario: simulate 30 days of operation, where each day has a small probability of an incident (Bernoulli distribution, $p = 0.008$/day, random downtime duration of 15–90 minutes); total downtime is computed and compared against the 99.5%/month uptime threshold.

```python
def simulate_availability(days=30, p_incident=0.008, seed=42):
    random.seed(seed)
    total_minutes = days * 24 * 60
    downtime = 0
    for _ in range(days):
        if random.random() < p_incident:
            downtime += random.uniform(15, 90)
    uptime_pct = 100 * (total_minutes - downtime) / total_minutes
    return {"uptime_pct": uptime_pct, "pass": uptime_pct >= 99.5}
```

Sample result: `uptime_pct` ≈ 99.85% ⇒ **PASS**.

4 Simulation Results Summary

NFR ID	Measured metric	Value	Conclusion
NFR-001	p95 latency (ms)	1233	PASS

NFR ID	Measured metric	Value	Conclusion
NFR-007	p99 conflict-detection time (s)	0.824	PASS
NFR-002	30-day uptime (%)	99.85	PASS

5 Self-assessment

Criterion	Max	Self-score	Notes
ISO 25010 classification	2.0	2.0	
NFR-001, NFR-007 simulation	2.5	2.5	
Self-chosen NFR + simulation	3.0	3.0	NFR-002, well analyzed
Presentation	2.5	2.5	
Total	**10.0**	**10.0**	